★ CS 401

Lec # 19-45
Chp No 7 To 17
*Final Term preparation
2024

Lec # 7
String Instructions

*String processing:-

A bit more powerful (than 8088) instruction that can process blocks of data in one go. They are called block processing or string instructions.

⇒ Block of memory, which is a video memory.

⇒ The vision of this memory for the processor is just a block of memory starting at a special address.

⇒ The five instructions are:- (Sh+Long MCAS. Img)

1- STOS:- Store String

2-LODS:- Load String
3-CMPS:- Compare String
4-SCAS:- Scan String
5-MOVS:- Move String.

> String instructions are complex instruction in that they perform a number of tasks against one instruction.

* Repeat while Equal (REPE)
* Repeat while Not Equal (REPNE)

⇒ Movs is used to move block of memory.

⇒ DF is imp in the case of overlapping blocks.

*Lec # 8
Software Interrupts

⇒ Interrupts in reality are events that accured outside the processor and the processor must be informed about them.

⇒ Interrupts are Asynchronous & unpredictable. → Quiz

Most Imp Topic

* The 8088 processor divides interrupts into two classes.
  1. Software Interrupt
  2. Hardware Interrupt
1. INT and IRET instruction formate and syntax.
2. The formation of IVT (interrupt vector table).
3. Operation of processor when interrupt in generated
* The interrupts reserved by processor and having special functions in 8088 are listed below:

- **INT 0:-** Division By Zero.
- **INT 1:-** Trap, Single Step Interrupt.
- **INT 2:-** NMI-Non Maskable interrupt.
- **INT 3:-** Debug Interrupt.
- **INT 4:-** Arithmetic Overflow, Change of Sign bit.

CamScanner

⇒ To book an interrupt we change vector corresponding to that interrupt.

⇒ BIOS and DOS interrupt.

• Basic input and Ouput Services (BIOS)

⇒ OS (operating system) Services are very high level and build upon the BIOS services.

⇒ BIOS services are very low level.

* Lec # 9
Real Time Interrupts and Hardware Interfacing

⇒ There is a single pin outside the processor called the INT pin that is used by external hardware to generate interrupts.

⇒ Programmable Interrupt Controller (PIC) ⇒ It has eight input signals and one output signal.

Quiz

⇒ PIC is an example of peripheral devices system.

For communicating with peripheral devices the processor uses **I/O Ports** → **Short+Quiz**

⇒ There are only Two operation with external world possible:
1. Read or
2. write

⇒ The **IN** and **OUT** instructions have a byte form and a word form but the byte forem is almost always used.

⇒ The soure registes in **OUT** and destination registes in **IN** is AL or AX, depending on which form is used.

⇒ programming Interrupt control has two ports **20** and **21.** → **Imp**

⇒ Port 20 is **Control port** while port 21 is athe **interrup mask registes** → which can be used for selectively enabling or disabling interrupts.

⇒ Each of the bits at port 21 corresponds to one of the **IRQ** lines.

* Lec # 10
Debug Interrupt

⇒ The three flags not used for mathematical operations are the:-

1. Direction flag
2. Interrupt flag
3. Trap flag

Short + Quiz

⇒ If the Interrupt flag is set, the after every instruction a type 1 interrupt will be automatically generated.

⇒ After every instruction the values of all registers are shown and the debugger waits for a key.

⇒ Another interrupt used by the debugger is the breakpoint interrupt INT 3.

⇒ INT 3 has a single byte opcode so it can replace any instruction.

* Most Imp Chp

*Lec # 11
Multitasking
* Concepts of Multitasking

⇒ To experience the power of assembly language we introduce how to implement multitasking.

⇒ We observed in the debugger that our thread of instructions was broken by the debugger. It got the control, used all registers, displayed and elaborate interface, waited for the key and then Restored processes ~~stored~~ State to what was immediately before interruption.

⇒ Our program resumed as if nothing happened.

⇒ The environment of a program in the very simple case is all its registers and stack.

⇒ The **IRQ 0** highest priority interrupt that is periodically coming to the processor.

⇒ The code to control this multitasking cannot be easily

C ++

written in a higher level language So we write it in assembly language.

⇒ The space where all registers of a task are stored is called Process control block or (PCB).

⇒ INT 08 that is saving and restoring the register is called the **Scheduler**.

⇒ And the whole event is called **context switch**.

Quiz + Short Q

⇒ When the program is executed the threads display the numbers independently.

⇒ As keys are pressed and new threads are registered, there is an obvious slowdown in the speed of multitasking.

⇒ To improve that, we can change the (Timer interrupt frequency) ⇒ Imp

* Lec # 12
Video Services

⇒ The Basic Inicsrupt Quipui System (BIOS) provide services for video, keyboard, serial port, parallel port, time etc.) -> Imp

⇒ The video services are exported as INT 10. -> Imp

* Video services are classified into two categories:
  1- Graphic mode services
  2- Text mode services

⇒ **In graphic mode:-**

a location in video memory corresponds to a dot on the screen.

⇒ **In Text mode:-**

This relation is not straight forward.

⇒ The video memory holds the ASCII of the character to be shown and the actual shape is read from a font definition stored else where is the memory.

=> Services of DOS are more cooked and at a higher level than BIOS. They provide less control but make routine tasks much easier.

=> The DOS input buffer has special formate where the **first byte** stores the **maximum character** buffer can holds, **second byte** holds the **number of characters** actually read on return.

* Lec # 13
Secondary Storage

• Physical Formation:

→ Short + Quiz

=> A floppy disk is a circular plate with a fine coating of magnetic material over it. the plate is enclosed in a plastic jacket. which has a cover that can slide to expose the magnetic surface.

⇒ If the head is fixed and the motor rotates the disk the readable area on the disk surface forms a circle called **track**.

⇒ Standard floppy disk has 80 tracks and 18 sectors per track with two head.

⇒ Hard disk have varying number of heads and tracks pertaining to their different capacities.

⇒ BIOS provides the simplest and most powerful interface to the storage medium.

⇒ **INT:-** 13 extensions that take linear 64 bit sectors number and handle the head, track, sector conversion themselves.

⇒ The physical sector 0 holds the master boot record and a partitio table toward end.

- **Storage Access Using DOS:-**

BIOS provides raw access to the storage medium while DOS gives a more logical view and more cooked services.

⇒ The actuall command tail is stored terminated by a CR → CR(Carriage Return)

⇒ Whenever a program is executed in DOS all available memory is allocated to it.

⇒ Memory must be freed using explicit calls to DOS for this purpose before a progam is executed.

⇒ **Device Drivers:-**

are operating system extensions that become part of the operating system and extend its services to new devices.

⇒ Device driver file starts with a header containing a link to the next driver in the first four bytes followed by device attribute word.

⇒ If the bit is zero the device is a character device and otherwise a block device.

⇒ Strategy routine is called whenever the device is needed and it is passed a request header. → Short + Quiz

⇒ For every command the first 13 bytes of request header are same. → Quiz

*LEC # 14.
*Serial Port Programming

⇒ Serial port is a way of communication among two devices just like the parallel port.

⇒ The data starts with a 1 bit called the start bit, one to two 0 bits called **stop bits.**

⇒ The duration of a bit must be precisely known at ~~the~~ both ends called the baud rate of the communication.

⇒ Every bit of line status conveys different information.

⇒ The port parameters in AL consist of the

- baud rate
- parity scheme
- number of stop bits.
- number of data bits.

⇒ Serial port is also accessible via I/O ports. COM1 is accessible via ports 3F8.

Imp Quiz

- **Serial Communication:-**

When two computers are connected using a serial cable.

* Lec # 15
Protected Mode Programming

⇒ 8088 architecture was a 16 bit processor.

⇒ Switching Processor in the never 32 bit mode is a very easy task.

- Just turn on the least significant bit of a new register called **CR0** (Control Register) and • The processor

Switches into 32 bit mode called **protected mode.**

⇒ All registers in 386 have been extended to 32 bits.

⇒ A 32 bit address register can access upto 4GB of memory. → Quiz

So memory access has increased a lot.

⇒ **The Table index (TI):-** is set to 0 to access the global table of descriptors, called the **GDT (Global descriptor table).**

⇒ The GDT itself is an array of descriptors where each descriptor is an 8 byte entry.

⇒ The base and limit of GDT is stored in a 48 bit register called the **GDTR.**

⇒ The first entry of the GDT must always be zero. called **null descriptor.** → **Jmp**

⇒ The 32 bit base in both descriptor is scattered into different places because of compatibility reasons.)

Short + Long

* The availability of larger registers and flexible addressing rules allows writing a much comprehensive version of the code.

⇒ VESA is the (Video Electronics Standard Association) VBE is the set of (Video BIOS Extensions) proposed by them.

Most Imp Topic

⇒ VESA VBE 2.0 standard include linear frame buffer mode that we will be using.

⇒ This mode allows direct access to the whole video memory.

⇒ Interrupt Handling:- in protected mode is also different.

⇒ instead of IVT at physical address 0 there is the IDT (interrupt descriptor table) located at physical address stored in IDTR, a special purpose register.

⇒ The IDTR is also a 48 bit register.

* Lecture No 16
Interfacing With High Level Language

⇒ To interface an assembly routine with a high level language program means to be able to call functions back & forth.

⇒ Two prevalent calling conventions are:-

1. C calling convention
2. Pascal calling convention

⇒ C prepends and underscore to every function or variable name

⇒ Pascal translate the name to

all uppercase.

⇒ In C parameters are pushed in reverse order with **rightmost** being pushed first.

⇒ In **Pascal** they are pushed in proper order with **leftmost** being pushed first.

⇒ In **NASM** symbol is declared with

1. → imported symbol:- External directive
2. → exported symbol:- Global directive

* Lec # 17
Comparison With other Processors.

* If assembly language has to be learned once and, every processor can be programmed by that person.

* **Two widely popular processors are:**

1. Motorolla 68K series (High performance embedded applications).
2. Sun SPARC (high end enterprise

→ Most Imp

* Intel x 86 series is known for its success in desktop market.

1. 68K processor have: 16 32bit general purpose registers named from A0-A7 and D0-D7.

- A0-A7 can holds addresses in indirect memory accesses.

⇒ Stack in 68K is not rigid( ) a structure as it is in x 86.

⇒ 8 bit condition code register (CCR) holds the X(Extend) N(Negative) Z(Zero) V(overflow) C(Carry) flags.

⇒ X is set to C for extended operations (addition, subtraction or shifting).

⇒ Motorolla processors allow bit addressing.

- A special MOVE 16 instruction also accepts a 16 byte block.
- Post increment or predecrement (part of memory access)

(An), (An)+, and -(An).

↓ ↓

Postincrement Predecrement

2. SUN SPARC: → Short + Quiz

is a very popular processing blonging to the RISC (reduced instruction set computer) family of processors.

- CISC (complex instruction set computer).

+ SPARC for (Scalabe Processor Architecture)

⇒ SPARC is a 64 bit processor.

- ASI (Address Space Identifier).

Quiz

⇒ There are 8 global registers and 8 alternate global register.. One of them is active at time and accessible as g0-g7.

⇒ The global registers can also be called r0-r7. → Short + Quiz

- in registers as r8-r15.
- Local registers as r16-r23
- Out registers as r24-r31.

Imp

* SPARC introduces a concept of register window.

=> One window is 24 registers and the active window is pointed to by a special register called **Current Window Pointer (CWP)**.

=> The register o6 is conventionally used as the stack pointer.

=> SPARC instructions have two sources and a distinct destination.